

CS 463/563 Final Project Journal

Student: Emmanuel Arthur

Project: Personal portfolio website (`emmarthur.github.io`)

Project links (required)

	URL
Deployed site	https://emmarthur.github.io/
GitHub repository	https://github.com/emmarthur/emmarthur.github.io
Video presentation	https://media.pdx.edu/media/t/1_cy2ey4w4

1. Project overview

I built a one-page professional portfolio site that meets the final project requirements: navbar, About (with photo), Previous Work, Projects (2–3 outside-course projects), and a Contact form. The site uses **HTML5**, **CSS**, **Bootstrap 5.3.3**, **JavaScript (ES6+)**, and **jQuery**.

The page is a single `index.html` file with section ids (`#home` , `#about` , `#previous-work` , `#projects` , `#contact`) so the navbar jumps to each area. I rebuilt the site in small steps with **8 meaningful commits** (the assignment requires at least 6).

Beyond-class components (additional requirement):

- **Bootstrap accordion** in Previous Work (`#workAccordion`)
- **Bootstrap carousel** in Projects (`#projectCarousel`)

These were not covered in the lab notebook; I learned them from the Bootstrap documentation.

2. How to reproduce this project

VS Code and Git are **not** required to run the site. It is static HTML; any modern browser is enough.

View online (no setup)

Open <https://emmarthur.github.io/> in Chrome, Edge, Firefox, or Safari.

Run from local files (no clone required)

With a local copy of the project folder (download, copy, or unzip):

1. Open the `emmarthur.github.io/` folder in File Explorer (Windows) or Finder (Mac).
2. Double-click `index.html` , or right-click → **Open with** → a modern browser.

Bootstrap and jQuery load from CDN links in the HTML, so connect to the internet for full styling and the carousel, accordion, and form validation. Do not move `index.html` out of the folder — `css/` , `js/` , and `images/` must stay in place.

Optional: clone from GitHub

```
git clone https://github.com/emmarthur/emmarthur.github.io.git
cd emmarthur.github.io
```

Then open `index.html` as above. Cloning is only needed for repo history or pushing changes.

Optional: VS Code + Live Server

For development only: open the folder in VS Code and use the Live Server extension. **Not required** to view or test the site.

File structure:

```
emmarthur.github.io/
  index.html           # One-page site (all sections)
  css/styles.css       # Custom theme and polish
  js/main.js           # Form validation, navbar active state
  js/jquery-interactions.js # jQuery section highlight
  images/              # Profile photo + project images
  README.md            # Run instructions and library list
```

3. Work log by commit

Each session below maps to a git commit. Messages match the repository history.

Commit 1 — Initial commit: empty starter files for personal website repo

What I did: Created the repo and folder layout: `index.html` , empty `css/styles.css` , empty `js/main.js` , empty `js/jquery-interactions.js` , and `images/` .

Why: Starting with structure before content makes later steps easier and matches how we organized homework in class.

Commit 2 — Link stylesheets and add navbar

What I did:

- Added HTML5 skeleton: doctype, `lang="en"`, charset, viewport meta, meta description, title.
- Linked Bootstrap 5.3.3 CSS (CDN) and `css/styles.css`.
- Built the navbar inside `<header>` with links to all five sections.
- Added Bootstrap navbar toggler for mobile (`data-bs-toggle` , `data-bs-target`).

Design decision: I moved `sticky-top` from `<nav>` to `<header>` so the whole top block stays fixed when scrolling (documented in `DESIGN-CHANGES.md`).

Issues: The hamburger menu did not open until Bootstrap JavaScript was added in a later commit.

Commit 3 — Add About and Previous Work sections

What I did:

- Added `<main id="main-content">` and a Home hero section (`#home`).
- Added About section with Bootstrap grid: photo (`images/1683647827149.jpg`) and bio text.
- Added Previous Work section with a **Bootstrap accordion** (`#workAccordion`) — three panels (two BlackRock roles + education).

Beyond class — accordion: Each panel uses `accordion-button` , `data-bs-toggle="collapse"` , and `data-bs-parent="#workAccordion"` so only one panel stays open at a time. I read the [Bootstrap Accordion docs](#) to learn the markup.

Commits 4 & 5 — Add Projects section and carousel

What I did:

- Added Projects section heading and intro.
- Built a **Bootstrap carousel** (`#projectCarousel`) with three slides, indicator dots, and prev/next controls.
- Added three project cards in a responsive grid (`#project-cards`).

Projects (all outside this course):

1. Retail Impact Simulator
2. StainCheck Mobile App
3. Web Security Exploitation Suite

Each card includes title, description, and GitHub link.

Beyond class — carousel: Structure follows [Bootstrap Carousel docs](#): `carousel-inner`, `carousel-item active`, `carousel-indicators`, and `data-bs-ride="carousel"`. Slideshow behavior required Bootstrap JS (added later).

Note: I have two commits with the same message here; the second added the contact section and script tags before a later commit split out `main.js`.

Commit 6 — Add contact form validation in `main.js`

What I did:

- Added Contact section with a labeled form (name, email, radio fieldset, message).
- Wrote `js/main.js` with custom validation (not browser default):
- Name: at least 2 characters
- Email: regex pattern check
- Message: at least 10 characters
- On failure: Bootstrap `is-invalid` class and text in `invalid-feedback` divs.
- On success: thank-you message in `#form-feedback`.
- Added navbar active-state updates when section hash changes.

Why novalidate on the form: So my JavaScript controls error messages instead of the browser's built-in popups.

Commit 7 — Add custom CSS, footer, jQuery, README, and project images

What I did:

- Filled in `css/styles.css`: teal theme, hero gradient, carousel caption overlay, form card styling, focus outlines.
 - Added site footer with copyright.
 - Added jQuery 3.7.1 (CDN) and `jquery-interactions.js` — brief orange outline on section when a nav link is clicked.
 - Wrote `README.md` with run instructions, libraries, and deploy URL.
 - Replaced placeholder picsum images with local Unsplash photos in `images/`.
-

Commit 8 — Documentation commits

What I did: Added `BUILD-STEPS.md`, `DESIGN-CHANGES.md`, and `VIDEO-PRESENTATION-OUTLINE.md` for my own build guide and video prep (not required for the live site, but useful for the journal and

presentation).

Commit 9 — Fix WAVE contrast and redundant link issues in navbar

What I did: Ran the [WAVE](#) extension and fixed first-round accessibility findings (see Section 5).

Follow-up (same session, may be a later commit): Fixed hero contrast by adding a `.hero-panel` with solid background `#003838` behind text on the gradient, per WAVE’s rule that text over gradients needs a defined background color.

4. Beyond-class feature for the video: Bootstrap carousel

The assignment requires at least one Bootstrap element **not covered in the lab notebook**. I used two (accordion + carousel). For the **video presentation**, I explain the **carousel** line by line — it has HTML structure, custom CSS, and relies on Bootstrap’s JavaScript bundle (no custom slide logic in `main.js`).

Also on the site (mention briefly in video): Bootstrap accordion in Previous Work (`#workAccordion`).

Bootstrap carousel (Projects) — video deep-dive

Location: `#projects` → `#projectCarousel`

Files to show:

File	What to explain
<code>index.html</code>	<code>#projectCarousel</code> wrapper, indicators, three <code>carousel-item</code> slides, prev/next controls, <code>data-bs-*</code> attributes
<code>css/styles.css</code>	Max width, image crop (<code>object-fit</code>), caption overlay background
<code>index.html</code> (bottom)	<code>bootstrap.bundle.min.js</code> — wires slide behavior from HTML attributes

How it works: Three `carousel-item` slides with project images and captions. Indicator dots and prev/next arrows use `data-bs-target="#projectCarousel"`. Bootstrap JS handles slide changes; I added CSS so captions stay readable over photos. The same three projects also appear as Bootstrap cards below for detail and GitHub links.

How I tested: After adding Bootstrap JS, I clicked arrows and dots and confirmed each slide appears. On mobile, captions hide (`d-none d-md-block`) so small screens show images only.

Bootstrap accordion (Previous Work) — brief mention only

Location: `#previous-work` → `#workAccordion`

One-line in video: accordion collapses job/education panels; learned from Bootstrap docs; not the line-by-line focus.

5. Issues encountered and how I fixed them

Issue	Cause	Fix
Navbar links low contrast (WAVE)	Bootstrap <code>navbar-dark</code> uses 55% opacity white on custom teal <code>#006060</code>	Set Bootstrap navbar CSS variables and link colors to <code>ffffff</code> in <code>styles.css</code>
Redundant Home links (WAVE)	Brand and “Home” nav both linked to <code>#home</code>	Removed “Home” nav item; brand name is the home link with <code>aria-current="page"</code>
Hero text low contrast (WAVE)	White text on CSS gradient without solid text background	Added <code>.hero-panel</code> with <code>background-color: #003838</code> on wrapper and text elements; 2 contrast errors still reported on final WAVE run
Redundant About links (WAVE)	Hero “Learn more” and nav “About” both went to <code>#about</code>	Changed hero CTA to a <code><button></code> that scrolls via <code>main.js</code>
Duplicate GitHub URLs (WAVE alert)	All three cards linked to same profile URL	Gave each project a distinct repo path on GitHub
Accordion/carousel not working	Bootstrap CSS only; no JS	Added <code>bootstrap.bundle.min.js</code> before <code>main.js</code>
Sticky nav overlap	Section headings hidden under navbar	<code>scroll-margin-top</code> on <code>section[id]</code> and <code>sticky-top</code> on <code><header></code>

Full WAVE notes are in `WAVE-AUDIT-FIXES.md` in the repo.

6. Accessibility audit results (WAVE)

I used the [WAVE](#) browser extension on the deployed site (<https://emmarthur.github.io/>) and on local Live Server. WAVE is required for the final project accessibility check.

First audit (before fixes)

WAVE finding	Count
Contrast errors	6
Alerts (redundant link)	2
Errors	0

Contrast errors (6) — navbar: Inactive nav links failed WCAG contrast against the custom teal background (`#006060`). Bootstrap's `navbar-dark` uses semi-transparent white (`rgba(255, 255, 255, 0.55)`), which does not meet 4.5:1 on our teal.

Alerts (2) — redundant Home links: The navbar brand (`Emmanuel Arthur`) and a separate “Home” nav item both linked to `#home` .

Fixes applied

Issue	Fix
Navbar contrast (6)	Set Bootstrap navbar CSS variables and link colors to full white (<code>#ffffff</code>) in <code>css/styles.css</code>
Redundant Home links (2)	Removed the “Home” nav item; brand name is the home link with <code>aria-current="page"</code> when on <code>#home</code>
Hero contrast (2)	Wrapped hero text in <code>.hero-panel</code> with solid background <code>#003838</code> on the panel, <code>h1</code> , and <code>.lead</code>
Redundant About links	Changed hero “Learn more about me” from <code></code> to a <code><button></code> that scrolls via <code>main.js</code>
Duplicate GitHub URLs	Gave each project card a distinct repo path instead of the same profile URL

Final audit (after fixes)

WAVE finding	Count
Contrast errors	2
Alerts (redundant link)	0
Errors	0

After pushing the navbar, hero panel, and redundant-link fixes, WAVE still reported **2 contrast errors** on the hero section — the `<h1>` and `.lead` paragraph over the gradient background. I added a

solid `#003838` background on `.hero-panel` and the text elements per WAVE's guidance for text on gradients, but the checker still flagged those two elements on the final run. Redundant-link alerts were cleared (0).

What I learned: Custom theme colors need to be re-checked when overriding Bootstrap defaults. WAVE's gradient rule is strict — even with a solid panel behind the text, the hero area can still fail depending on how the checker reads layered backgrounds.

Other accessibility practices

- **Semantic HTML:** `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<figure>`, form labels with `for` attributes, `fieldset/legend` for radio group.
- **ARIA:** `aria-label` on nav and carousel, `aria-labelledby` on sections, `aria-current` on active nav/brand, `aria-live="polite"` on form feedback.
- **Keyboard / focus:** Visible focus outlines in custom CSS; navbar and form controls are keyboard reachable.
- **Formatting:** Ran Prettier on HTML, CSS, and JavaScript before commits.

7. Deployment

1. Created public repo `emmarthur.github.io` on GitHub.
2. Pushed `main` branch after each meaningful commit.
3. Enabled GitHub Pages: **Settings** → **Pages** → **Source: main, folder / (root)**.
4. Live site: <https://emmarthur.github.io/>

8. Outside sources and libraries

Resource	How I used it
Bootstrap 5.3 Docs	Navbar, grid, accordion, carousel, form classes
Bootstrap Accordion component	Previous Work section
Bootstrap Carousel component	Projects slideshow
jQuery 3.7.1	CDN + <code>\$(document).ready</code> , <code>.on()</code> , <code>.addClass()</code> in <code>jquery-interactions.js</code>
jsDelivr CDN	Bootstrap CSS/JS hosting
WAVE Web Accessibility Evaluation Tool	Automated accessibility audit
Unsplash (Unsplash License)	Project images — retail checkout, coffee spill, circuit board

Resource	How I used it
CS 463/563 lab notebook	HTML forms, Bootstrap basics, accessibility unit patterns
Prettier	Code formatting

No premade themes or copied portfolio templates were used. Bio and job content are my own.

9. What breaks if parts are removed

The portfolio is a single page with linked assets. Removing a piece changes what loads or what still responds to clicks.

Removed piece	Effect on the live page
<code>css/styles.css</code>	Teal theme, hero panel, carousel caption styling, navbar WAVE contrast fixes, and focus outlines disappear.
<code>js/main.js</code>	Custom contact validation, success alert, navbar active state, and hero scroll-to-About stop working.
<code>js/jquery-interactions.js</code>	Orange section highlight on nav clicks stops.
<code>bootstrap.bundle.min.js</code>	Accordion, carousel, and mobile menu clicks do nothing.
<code>#projectCarousel</code>	Beyond-class carousel is gone; project cards may remain below.
Accordion in Previous Work	Long static job history instead of collapsible panels.
WAVE fixes documented in Section 6	Contrast and redundant-link findings return; see <code>WAVE-AUDIT-FIXES.md</code> .
<code>images/</code> files	Broken images in About and Projects.

Detailed notes per build step: `BUILD-STEPS.md`. Carousel-specific impact: `CAROUSEL-DEEP-DIVE.md`.

10. Video presentation note

A separate **10–12 minute video** walks through these commits and demos the live site. The recording is hosted on PSU Media Space:

https://media.pdx.edu/media/t/1_cy2ey4w4

Outline and talking points: `VIDEO-PRESENTATION-OUTLINE.md`. The video goes into more detail on commits where this journal is shorter.